Roll No.: BL.AI.U4.AI02AI40

Amrita Vishwa Vidyapeetham
School of Artificial Intelligence, Faridabad
B.Tech Mid-Term Examinations – October 2025
First Semester
Artificial Intelligence & Data Science

25AIM102 Computational cell Anatomy and Physiology

Duration: Two hours
Maximum: 50 Marks

Course Outcomes (COs):

CO1: Identify and describe major human anatomical structures and physiological processes using digital anatomy tools.

CO2: Use simulation platforms to model and visualize cell, tissue, and organ-level functions.

CO3: Interpret simple biomedical signals (EEG, ECG, PCG) using open datasets.

CO4: Demonstrate understanding of the relationship between computational models and real physiology through guided projects.

Answer all the Questions. Each Question carries 5 Marks. Illustrate with appropriate diagrams wherever necessary.

1. Explain the structural organization of the human skeletal system. How can AI-based medical imaging tools (e.g., CNNs, segmentation models) help in detecting fractures, osteoporosis, or skeletal deformities? [5][CO01][BTL 2]

2. Describe the major types of muscles and their roles in body movement. How can AI-driven biomechanical simulations assist in rehabilitation engineering and prosthetic limb design? [5][CO01][BTL 2]

3. Outline the organs and functions of the digestive system. How do AI-powered endoscopic vision models assist in early detection of gastric and intestinal diseases? [5][CO02][BTL 2]

4. Name all the anatomical planes and explain their significance in developing advanced automation using AI and DS Technology. [5][CO02][BTL 2]

5. Explain the anatomy and physiology of the human excretory system. How can AI algorithms be used to detect kidney stones and assess renal function using ultrasound? [5][CO02][BTL 2]

6. Describe how IoT-enabled devices combined with data analytics can monitor urinary output and electrolyte balance in patients with renal failure. [5][CO02][BTL 2]
7. Explain the functional components of the central (CNS) and peripheral nervous systems (PNS). How could AI and deep learning models be utilized in brain imaging for diagnosing neurological disorders? [5][CO01][BTL 2]
8. Discuss how AI-based brain–computer interfaces (BCIs) can help restore movement in paralyzed patients and support neuroprosthetic development. [5][CO01][BTL 2]
9. Describe the principles of hemodialysis and peritoneal dialysis. How can AI and predictive analytics optimize dialysis parameters like urine filtration rate and fluid management? [5][CO02][BTL 2]
10. Explain how machine learning models can analyze patient data to predict dialysis complications and personalize treatment schedules. [5][CO02][BTL 2]

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Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO01	**25**	BTL 1	-
CO02	**25**	BTL 2	**50**
CO03	-	BTL 3	-
CO04	-	BTL 4	-